

Two-Photon Absorption by Fluorene Derivatives: Systematic Molecular Design

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In this article, we employ a systematic approach to the computational quantum chemical study of the two-photon absorption (2PA) properties of 161 representative molecules containing a symmetrically substituted fluorene unit. The molecules studied contain *meta*- or *para*-substituted phenyl groups, five- and six-membered heterocycles, and benzo derivatives of five-membered heterocycles. The computational procedure employed to calculate the 2PA parameters was previously described [*Chem. Mater.* **2008**, *20*, 4142] and is based on semiempirical electronic structure methods: the RM1 model to optimize the molecular geometry and the INDO/S method to calculate the spectroscopic properties of the molecules. We further advance a new, simplified expression employed to calculate an approximate three-level contribution of the imaginary part of the negative component of the second hyperpolarizability. We then show that, in order to rationalize the 2PA cross sections for the substituted fluorenes, the three-level approximation has to be adapted to include a fourth state. That done, we advance that the parameter most responsible for the large observed variation in the calculated values of the 2PA cross sections for the substituted fluorenes is the effective transition dipole moment between the 1PA-active state and the two 2PA-active states. Based on our results, we discuss three structural effects that can contribute to the value of the 2PA cross section and show how they can be tuned. We conclude by propositioning novel putative molecules with potentially large values of 2PA cross sections, such as 2,2'-(9,9-dialkyl-9H-fluorene-2,7-diyl)dibenzo[d]oxazole.

1. Introduction

Two-photon absorption (2PA) process is a nonlinear optics phenomenon predicted theoretically by Maria Göppert-Mayer in 1931.¹ In 2PA, two photons, with equal or different energies, combine to yield an excitation of the molecule which equals the sum of the energies of the two photons taken separately. The first experimental measurement of the 2PA process took place only in 1961,² after the appearance of laser sources. Different from other nonlinear optical processes, such as second harmonic generation, which to be observed requires materials without inversion symmetry, the 2PA process can be observed in isotropic materials, such as liquids. The 2PA process has selection rules that are different from those of the more usual one-photon absorption (1PA) processes. While the 1PA processes require ground and excited states with different parities to occur, the 2PA process can only happen between states of the same parity in materials with inversion symmetry. Therefore, in such centrosymmetric materials, the 1PA and 2PA processes access different excited states.

There are two important properties that set the 2PA process apart from the customary 1PA processes. First, the two photons absorbed have wavelengths where the absorption of one photon is very weak, meaning that the incident photons are able to penetrate deeper into the sample. Second, the probability of absorption of two photons scales with the square of the intensity of the incident laser, implying that the absorption of two photons is localized around the focus of the incident laser beam. Together, these two properties result in greater spatial resolution for 2PA processes, when compared to the more usual 1PA processes.

The absorption of two photons by organic molecules has several technologically important applications, such as fluorescence microscopy;^{3–8} 3D microfabrication;^{9–13} optical data storage;^{10,14–16} photodynamic therapy;^{17–19} and optical limiting.^{20–24} Recently, uses of organic molecules that absorb two photons, such as nanoscopic sensors of metals,^{25–29} pH,^{30–32} or fluoride anions,^{33–36} started to emerge. Because of all these important applications, the theoretical design and the synthesis of new organic molecules with large values for the cross section for the absorption of two photons, $\delta(\omega)$, are much needed endeavors.

A common strategy employed in the design of organic molecules with large values of $\delta(\omega)$ is the linkage of two donor (D) or acceptor (A) groups through a conjugated bridge (B) yielding molecules with the donor-bridge-donor (D-B-D) or acceptor-bridge-acceptor (A-B-A) arrangements.^{37–43} Upon excitation, these linear symmetric quadrupolar systems exhibit transference of charge between the extremities and the center of the molecule.

In two different articles,^{44,45} Kuzyk studied the maximum fundamental limits for the values of $\delta(\omega)$. They found that, for all quadrupolar organic molecules analyzed, the experimentally observed values of $\delta(\omega)$ are at least 2 orders of magnitude smaller than these fundamental limits. This observation means that there is still much room to design new molecules possessing values of $\delta(\omega)$ that are much larger than the ones already measured.

In a previous article,⁴⁶ we began to explore such possibilities through the study of quadrupolar arrangements of two identical type A mesoionic rings connected through polyenic bridges. In that article, we employed a systematic approach to construct several hundred distinct molecules that were submitted to semiempirical electronic structure calculations with the objective of ordering the molecules as a function of increasing values of $\delta(\omega)$. The results showed that quadrupolar arrangements

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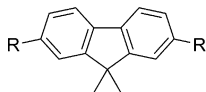


Figure 1. Structure of the symmetrically substituted 9,9-dimethylfluorenes studied in the present article, where R stands for 161 different groups.

containing type A mesoionic rings in their structure are very promising candidates for systems with large values of $\delta(\omega)$.

In the present study, we use the computational procedure proposed in our previous article⁴⁶ to study the 2PA properties of a series of symmetrically substituted 9,9-dimethylfluorenes (Figure 1). We chose to employ the fluorene unit as the conjugated bridge in our study because of the growing importance that this unit has in the design of two-photon absorbing materials.

In section 2, we describe the procedure employed in our calculations. In section 3, we present and discuss the results for our calculations for the symmetrically substituted fluorenes, and in section 4, we present the main conclusions of this work, including several putative molecules with potentially large values of $\delta(\omega)$.

2. Computational Procedure

The cross section for the absorption of two photons of frequency ω is given by

$$\delta(\omega) = a \frac{\pi^2 \hbar \omega^2}{n^2 c^2} L^4 \text{Im}\langle \gamma(-\omega; \omega, \omega, -\omega) \rangle \quad (1)$$

where n is the refractive index of the medium, c is the speed of light, and L is a local field factor. In our calculations, we consider the molecules in a vacuum, and therefore, the values of n and L are taken to be equal to one. In eq 1,

$$\gamma_{ijkl}^{2P} = \frac{1}{6\hbar^3} \sum_{o,p,q \neq f} \left(\frac{((\omega_{of} - \omega + \sqrt{-1}\Gamma)\mu_{fo}^i \bar{\mu}_{op}^l + (\omega_{of} - \omega - \sqrt{-1}\Gamma)\mu_{fo}^l \bar{\mu}_{op}^i)(\bar{\mu}_{pq}^k \mu_{qf}^j + \bar{\mu}_{pq}^j \mu_{qf}^k)}{(\Gamma^2 + (\omega_{of} - \omega)^2)(\omega_{pf} - 2\omega - \sqrt{-1}\Gamma)(\omega_{qf} - \omega - \sqrt{-1}\Gamma)} \right) \quad (2)$$

and a simplified, but still exact, form of γ^N given by

$$\gamma_{ijkl}^N = \frac{1}{6\hbar^3} \sum_{o,q \neq f} \mu_{fo}^i \left[\frac{4(\Gamma^2 + \omega^2 - \omega_{of}^2)(\omega_{of} + \omega_{qf})}{((\Gamma^2 + \omega^2)^2 + 2(\Gamma^2 - \omega^2)\omega_{of}^2 + \omega_{of}^4)((\Gamma - \sqrt{-1}\omega)^2 + \omega_{of}^2)} \mu_{fo}^l \mu_{fq}^j \mu_{fq}^k + \right. \\ \left. \frac{4[\omega_{qf}(\Gamma^2 - \omega^2 + \omega_{qf}^2)((\Gamma - \sqrt{-1}\omega)^2 - \omega_{of}^2) + \omega_{of}(\Gamma^2 + \omega^2 + \omega_{qf}^2)((\Gamma^2 + \omega^2) - \omega_{of}^2)]}{((\Gamma^2 + \omega^2)^2 + 2(\Gamma^2 - \omega^2)\omega_{qf}^2 + \omega_{qf}^4)((\Gamma - \sqrt{-1}\omega)^2 + \omega_{of}^2)^2} (\mu_{fo}^k \mu_{fq}^j + \mu_{fo}^j \mu_{fq}^k) \mu_{fq}^l \right] \quad (4)$$

In eqs 3 and 4, f is the ground state, and o , p , and q are excited states of the molecule. The transition dipole moment in the direction i , between states o and p , is given by μ_{op}^i , and $\bar{\mu}_{op}^i = \mu_{op}^i - \mu_{fp}^i \delta_{op}$ is the difference between the permanent dipole moments of states o and f . The transition energy between states f and o is given by $E_{of} = \hbar\omega_{of}$, and Γ is the damping factor for the excited states. In our calculations, we assume the same value of $\hbar\Gamma = 0.1$ eV for all excited states. It is important to stress that eqs 3 and 4 were obtained using the perturbative definition of the hyperpolarizabilities.⁴⁷ In our program, in order to calculate $\delta(\omega)$, we implemented only the imaginary parts of eqs 3 and 4.

To calculate the imaginary part of the hyperpolarizability using eqs 3 and 4, we need to obtain values for μ_{op}^i and E_{of} using quantum chemical methods. We chose to employ semiempirical electronic structure methods in our calculations. This choice was motivated by

$\text{Im}\langle \gamma(-\omega; \omega, \omega, -\omega) \rangle$ is the imaginary part of the second dynamic hyperpolarizability of the molecule. The value of the numerical constant a depends on the convention employed to define the hyperpolarizability γ , for which there are several conventions which have been described by Willetts et al.⁴⁷ The two most important ones are the Taylor series expansion and the perturbative conventions. When γ is defined using a Taylor series expansion, the constant a is equal to 4; and when γ is defined using the perturbative convention, the constant a is equal to 24. In turn, the value of γ in the Taylor series convention is six times the value of γ in the perturbative convention, rendering the value of $\delta(\omega)$ identical in both cases, as thoroughly explained in a recent review by Terenziani et al.⁴⁸ In this article, we employ the perturbative definition of γ .

The values of $\delta(\omega)$ observed experimentally result from a spherical average of γ , given by

$$\langle \gamma \rangle = \frac{1}{5} \left[\sum_i \gamma_{iiii} + \frac{1}{3} \sum_{j \neq i} (\gamma_{ijij} + \gamma_{ijji} + \gamma_{jiij}) \right] \quad (2)$$

where $i, j = x, y, z$. The Cartesian components of the hyperpolarizability, γ_{ijkl} , are obtained using a sum over states (SOS) expression proposed by Orr and Ward (eq 43c of their article).⁴⁹ To derive their expression, Orr and Ward employed time-dependent perturbation theory within the electric dipole approximation to model the interaction between the molecule and the electric field of the incident laser. For our purposes, their expression can be written as $\gamma_{ijkl} = \gamma_{ijkl}^{2P} + \gamma_{ijkl}^N$, where γ^{2P} contains all the terms responsible for producing the maxima of absorption of two photons, together with a few extra terms; and γ^N contains what is usually called the negative component of γ .

As detailed in our previous article,⁴⁶ in our calculations of $\delta(\omega)$ we employ a truncated version of γ^{2P} given by

the fact that ab initio electronic structure methods are much more demanding computationally than the semiempirical ones. Therefore, using ab initio methods, we cannot perform, at a low computational cost, systematic studies, where we calculate large numbers of molecules, on the order of hundreds. However, using semiempirical methods we can perform such studies in a reasonable amount of time and still get useful results.

To calculate the properties of the excited states of our molecules, we performed configuration interaction (CI) calculations with the semiempirical INDO/S⁵⁰ Hamiltonian in the ZINDO⁵¹ program. In our previous article,⁴⁶ we showed that including only single excitations in the CI (CIS), with an excitation window that encompasses all occupied and all unoccupied molecular orbitals, we are capable of producing values for the cross section in the maximum of the 2PA peak,

$\delta_{\max.}$, useful for molecular design, while still obtaining accurate values for both 1PA and 2PA wavelengths (λ_{1pa} and λ_{2pa}). We showed that this semiempirical CIS procedure is capable of ordering homologous molecules in terms of increasing values of $\delta_{\max.}$, and that the procedure, which also includes double excitations in the INDO/S calculations, yields unreliable values for both λ_{1pa} and λ_{2pa} .⁴⁶ This is because the semiempirical INDO/S Hamiltonian was parametrized to predict experimental values of properties of electronic excited states using CI calculations that include single excitations only. The inclusion of double excitations in the INDO/S calculations will unbalance the INDO/S semiempirical parametrization, yielding results, mainly excitation energies, which may significantly deviate from the experimental ones. We would like to emphasize that this single excitations only approach⁴⁶ is solely applicable to INDO/S calculations, and does not imply that double excitations are either phenomenologically unimportant or not required for the ab initio determination of 2PA cross sections.

As indicated above, the excitation window used in this article includes all occupied and all unoccupied molecular orbitals in the CIS calculations. Of all the excited states generated in the CIS calculations, we include in our calculations of the 2PA cross section only the 100 lowest excited states in eqs 3 and 4. We obtained the values of $\delta_{\max.}$ in the Göppert-Mayer unit, GM, which is equal to 10^{-50} cm⁴ s/photon.

In the present work, we optimized the geometries of our molecules using the semiempirical RM1 (Recife Model 1) Hamiltonian⁵² implemented in the MOPAC 2009 program.⁵³ The RM1 methodology was already parametrized for all the atoms we needed for this study and yields optimized geometries that compare better with experiment than either AM1⁵⁴ or PM3⁵⁵ methodologies.

In principle, the values of the 2PA cross sections are a complicated function of the excitation energies, the transition dipole moments, and the differences in the permanent dipole moments between the ground and excited states. A technique that has been shown to be useful in the interpretation of nonlinear optical properties, such as 2PA cross sections, is the so-called few-level approximations,^{56,57} particularly the two- and three-level approximations. Recently, the three-level approximation has also been extended to include contributions from electron-vibration coupling and interaction with solvents to the 2PA cross sections of fluorene-based quadrupolar molecules.⁵⁸ Using eq 3 as a starting point, it is possible to obtain an approximate three-level expression for the imaginary part of γ^{2P} . This three-level approximation is obtained by assuming that the molecule (a) is linear, and we need only the γ_{xxx} component of the hyperpolarizability, and (b) possesses only one excited state with sizable transition dipole moments from both the ground and 2PA-active excited states. In our notation, this intermediate state is called *o*, making $q = o$ in eq 3. In the notation of eq 3, the state that absorbs two photons is given by the index *p* and the frequency of the two absorbed photons is given by $\omega^{2P} = (1/2)\omega_{pf}$. In order to obtain the cross section in the maximum of the 2PA peak, we make the substitution $\omega = \omega^{2P}$ in the resulting equation. The final step is to assume that $4\Gamma^2 \ll (2\omega_{of} - \omega_{pf})^2$ in the denominator and finally obtain the three-level approximation given by

$$\text{Im } \gamma_{xxx}^{3L} = \frac{8}{3\hbar^3} \frac{(\mu_{jo}^x \mu_{op}^x)^2}{\Gamma(2\omega_{of} - \omega_{pf})^2} = \frac{8}{3\hbar\Gamma} \frac{(\mu_{jo}^x \mu_{op}^x)^2}{(2E_{of} - E_{pf})^2} \quad (5)$$

It is important to reiterate that eq 5 is appropriate to describe cases where the direct excitation to the 2PA excited state is not

TABLE 1: Comparison between the Values of the Negative Correction to the Three-Level 2PA Cross Section Calculated from Eqs 6 (δ^{neg}) and 7 ($\delta_{\text{approx}}^{\text{neg}}$)^a

E^{2P} (eV)	δ^{neg} (GM)	$\delta_{\text{approx}}^{\text{neg}}$ (GM)	ΔE (eV)
2.00	−1	−1	1.75
2.25	−3	−3	1.50
2.50	−8	−7	1.25
2.75	−23	−20	1.00
3.00	−82	−74	0.75
3.25	−440	−410	0.50
3.50	−5701	−5483	0.25

^a As a function of the energy of the two photons absorbed ($E^{2P} = (1/2)\hbar\omega_{pf}$). ΔE is the distance between the 1PA and 2PA peaks (half the detuning factor). The calculated values are rounded to the nearest integer. These values were obtained using $E_{of} = \hbar\omega_{of}$ equal to 3.75 eV and μ_{jo}^x equal to 9D in eqs 6 and 7. (1 GM = 10^{-50} cm⁴ s/photon).

permitted by symmetry ($\mu_{pp}^x \approx 0$). For the case of molecules of lower symmetry, it is essential to also include in the analysis the direct excitation to the 2PA state that is also 1PA active. This gives rise to the two-level approximation, where there is a contribution proportional to the square of the product between μ_{pp}^x and the change in the permanent dipole moment ($\Delta\mu_{pp}^x$) between the ground state and the 2PA excited state. The molecules studied in the present article are only approximately symmetric and possess 1PA active excited states in a region where Ti:sapphire laser-facilitated 2PA excitations are possible. In the results that follow, we will focus on the region of the maximum 2PA cross section. We verified that, in this region, the relevant excited states either have direct transition dipoles from the ground state that are small or show virtually no change in the permanent dipole moment upon excitation and, therefore, are not susceptible to a direct 2PA transition. Based on this result, we ignored the contribution due to the two-level approximation terms and use only eq 5 in our analysis of the approximate 2PA cross sections.

Within the linear molecule assumption, we have that eq 2 leads to $\text{Im } \langle \gamma \rangle = (1/5)\text{Im } \gamma_{xxx}^{3L}$. Using this approximate value of $\text{Im } \langle \gamma \rangle$ in eq 1, with $\omega = \omega^{2P}$, we obtain an approximate three-level value for the 2PA cross section, δ^{3L} . Equation 5 shows that, in order to increase the value of δ^{3L} , we may either increase the values of the transition dipole moments μ_{jo}^x and μ_{op}^x or decrease the value of the detuning factor $E_{\text{det}} = 2E_{of} - E_{pf}$. It is important to stress that the final step employed to obtain eq 5 restricts the validity of the three-level approximation to situations where we are far from the double resonance condition $\omega_{of} = (1/2)\omega_{pf}$. Besides the erroneous divergence of δ^{3L} close to the double resonance condition, there are also practical reasons to avoid such situation because, when $(1/2)\omega_{pf}$ is close to ω_{of} , the much stronger 1PA peak overshadows the weaker 2PA peak, rendering the molecule uninteresting for 2PA applications.

It is worth mentioning that, in general, the three-level approximation works very well for linear quadrupolar molecules. Exceptions to this rule are the molecules containing type A mesoionic rings that were studied in our previous article.⁴⁶ For such systems, the values of the detuning factors were not sufficiently large in order to fully justify the assumption $4\Gamma^2 \ll (2\omega_{of} - \omega_{pf})^2$.

To obtain the approximation in eq 5, we completely ignored the contribution of the negative component to the imaginary part of γ . It can be shown that, as the detuning factor decreases, the contribution of the negative component of the hyperpolarizability to $\delta(\omega)$ becomes important. In our previous article,⁴⁶

we have proposed that, by using the same assumptions employed to obtain $\text{Im } \gamma_{xxxx}^{3L}$, that is, (a) only the x component of the hyperpolarizability contributes, (b) only one dominant excited state o absorbs one photon, and (c) $\omega = (1/2)\omega_{pf}$, we can obtain the following semixact three-level expression for the imaginary part of the negative component of γ at the position of the maximum of the 2PA peak

$$\text{Im } \gamma_{xxxx}^{\text{neg}} = \frac{512(\mu_{fo}^x)^4 \Gamma \omega_{of} \omega_{pf}}{3\hbar^3} \times \frac{(64(2\Gamma^2 - 3\omega_{of}^2)(\Gamma^2 + \omega_{of}^2)^2 + 16(3\Gamma^4 + 4\Gamma^2\omega_{of}^2 + 5\omega_{of}^4)\omega_{pf}^2 - (4\omega_{of}^2 + \omega_{pf}^2)\omega_{pf}^4)}{(16(\Gamma^2 + \omega_{of}^2)^2 + 8(\Gamma^2 - \omega_{of}^2)\omega_{pf}^2 + \omega_{pf}^4)^3} \quad (6)$$

In the present article, instead of using the semixact expression above to calculate an approximate three-level contribution for the imaginary part of the negative component of γ , we advance, for the first time, an approximate expression for $\text{Im } \gamma_{xxxx}^{\text{neg}}$ given by

$$\text{Im } \gamma_{xxxx}^{\text{approx,neg}} = -\frac{64}{3\hbar^3} \frac{(2\omega_{of} - \omega_{pf})^2 (\mu_{fo}^x)^4 \Gamma}{(4\Gamma^2 + (2\omega_{of} - \omega_{pf})^2)^3} \quad (7)$$

This expression is much simpler and easier to interpret than eq 6. Equation 7 can be obtained from eq 6 through three simple steps: first, we make the substitution $\omega_{pf} = 2\omega_{of} - \varepsilon$; then, we take the limit of the resulting equation for ω_{of} going to infinity; and, finally, we undo the substitution of the first step, making $\varepsilon = 2\omega_{of} - \omega_{pf}$ to obtain eq 7.

We showed in our previous article⁴⁶ that we can use the results of eq 6, together with eqs 1 and 2, in order to obtain a semixact negative correction, δ_{neg} , to the approximate three-level 2PA crosssection δ^{3L} . Similarly, we can use eq 7 in eqs 1 and 2 to obtain an approximate negative correction, $\delta_{\text{approx}}^{\text{neg}}$. In Table 1, we compare some values of the δ_{neg} and $\delta_{\text{approx}}^{\text{neg}}$ obtained using some typical values of $E_{of} = \hbar\omega_{of}$ and μ_{fo}^x for several values of the energy of the two photons absorbed $E^{2P} = (1/2)\hbar\omega_{pf}$. We can observe in Table 1 that, only when the distance between the 1PA and 2PA peaks becomes very small, there is a noticeable difference between the values of the negative corrections obtained from eqs 6 and 7. Therefore, these results show that eq 7 is a valid approximation for eq 6. We can observe in Table 1 that, as the detuning factor decreases, the magnitude of the negative correction increases.

3. Results and Discussion

For the present article, we employed a systematic approach to study the 2PA properties of 161 distinct molecules (Figure 1). The R groups chosen were *meta*- or *para*-substituted phenyl groups, five- and six-membered heterocycles, and benzo derivatives of five-membered heterocycles.

In our calculations, we observed that the most promising molecule has a calculated value of δ_{max} more than six times larger when compared to the least promising molecule. To explain this variation, we applied an adapted version of the three-level approximation (eq 5) to our molecules. The three-level approximation assumes one 1PA-active excited state o and one 2PA-active excited state p . In the case of several of our molecules, we observed two or more nearly degenerate

2PA-active excited states. As an example, in Figure 2, we have the energy level diagram of the excited states involved in the 2PA process for the molecule with R equal to phenyl. We can observe in Figure 2 that we have sizable values of transition dipole moments between the two nearly degenerate 2PA excited states p and p' and the 1PA-active excited state o . To take this into account, we introduce a fourth state in

the three-level approximation and define an effective value for the transition dipole moment between the 1PA-active and the two 2PA-active excited states given by

$$\mu_{op}^{\text{eff}} = \left(\sum_{p'=1}^2 \mu_{op'}^2 \right)^{1/2} \quad (8)$$

The definition above follows directly from eq 5 after we take a common value for the detuning factor (E_{det}) for all the 2PA-active excited states. Before using eq 8, we projected the transition dipole moments $\mu_{op'}$ in the direction of μ_{fo} . This basically means that we defined our x axis for the approximate calculation of the 2PA cross section parallel to the 1PA transition dipole moment μ_{fo} . In Table S1 of the Supporting Information, we have the three-level approximation analysis for all molecules considered in the present work. For these molecules, the values of the detuning factors are not sufficiently small in order to render the three-level approximation unreliable (E_{det} varies between 1.236 and 2.366 eV for these molecules). For this same reason, we can observe in Table S1 that the negative correction ($\delta_{\text{approx}}^{\text{neg}}$) to the three-level 2PA cross section δ^{3L} is, in general, small, being on average less than 2% of δ_{max} . Only in five cases is the value of $\delta_{\text{approx}}^{\text{neg}}$ more than 5% of δ_{max} . Therefore, we are not including the values of $\delta_{\text{approx}}^{\text{neg}}$ in the analysis that follows.

In Figure 3, we have a comparison between the values of δ_{max} calculated using 100 excited states in eqs 3 and 4 and the approximate values δ^{3L} obtained from eq 5, for all molecules considered in this article. We can observe that, except for a few atypical molecules, the agreement between δ^{3L} and δ_{max} is excellent. The three-level approximated results are on average 6.8% smaller than δ_{max} . The R groups of the two atypical

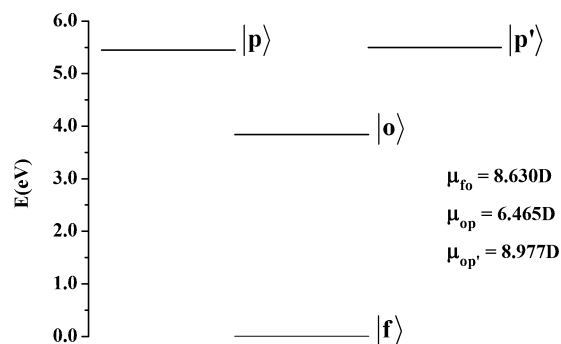


Figure 2. Energy level diagram of the excited states involved in the 2PA process for the molecule with R equal to phenyl. The 1PA-active excited state is called o and the two 2PA-active excited states are called p and p' . In the diagram, we also have the transition dipole moments involved in the 2PA process. For this molecule, we have $\mu_{op}^{\text{eff}} = 11.062$ D.

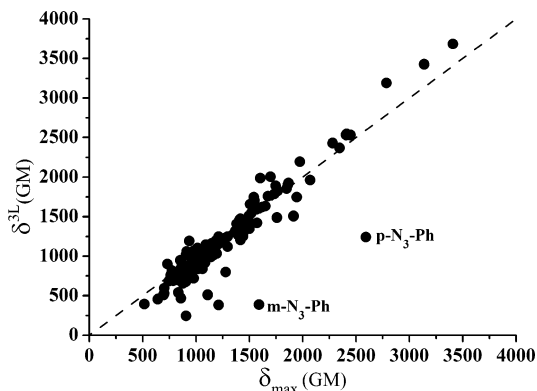


Figure 3. Comparison between the 2PA cross section calculated using 100 excited states in eqs 3 and 4 ($\delta_{\max}$) and the 2PA cross section obtained using the three-level approximation in eq 5 (δ^{3L}) for all the molecules studied in the present work. The dashed line has slope of 1. Also identified are two of the molecules where the discrepancies between δ^{3L} and $\delta_{\max}$ are the largest. (1 GM = 10^{-50} cm⁴ s/photon.)

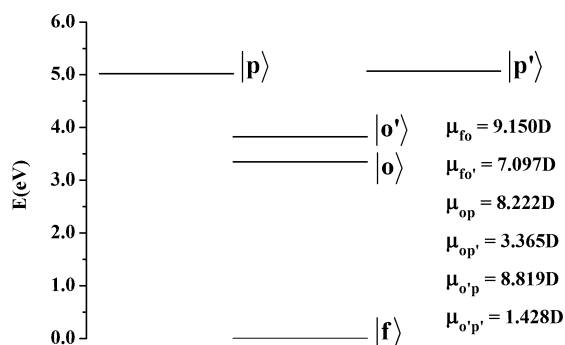


Figure 4. Energy level diagram of the excited states involved in the 2PA process for the molecule with R equal to *para*-azido-phenyl. The two 1PA-active excited states are called *o* and *o'*, and the two 2PA-active excited states are called *p* and *p'*. In the diagram, we also have the transition dipole moments involved in the 2PA process.

molecules that have the largest absolute errors are indicated in Figure 3. We have that, in both cases, the three-level approximated results grossly underestimate the value of the 2PA cross-section and that both molecules possess phenyl groups with the azido substituent either in the *para* or in the *meta* positions. In Figure 4, we have the energy level diagram of the excited states involved in the 2PA process for the molecule containing the *para*-azido-phenyl group. We can observe that, besides the two nearly degenerate 2PA-active excited states *p* and *p'*, we also have two 1PA-active excited states *o* and *o'*, that are not degenerate, with sizable values for the transition dipole moments. The presence of more than one 1PA-active excited state contributing to the value of the 2PA cross section $\delta_{\max}$ renders the three-level approximation inadequate, and consequently, the value of δ^{3L} underestimates $\delta_{\max}$. A similar arrangement of excited states is also observed for the molecule containing the *meta*-azido-phenyl group, as well as for a few other molecules.

Substituting eq 5 in eqs 1 and 2, we have that the value of δ^{3L} is given by

$$\delta^{3L} = k \frac{E_{pf}^2 \mu_{fo}^2 \mu_{op}^2}{E_{\text{det}}^2} \quad (9)$$

where we have collected all the numerical constants in *k* and μ_{op} is the effective value given by eq 8. We have that the value

of δ^{3L} depends on four parameters (E_{pf} , E_{det} , μ_{fo} , and μ_{op}). We would like to find out the role played by each one of these four parameters in the observed variation of δ^{3L} . In order to do so, we recalculated the values of δ^{3L} four times, keeping each time one of the parameters constant at its average value, and varying the other three for each molecule. This procedure works as follows. By fixing one parameter at its average value, we remove its influence from the overall variation of δ^{3L} . If this removal does not affect the general trend of the 2PA cross section, we may conclude that the effect of this fixed parameter on the overall ordering of δ^{3L} across the compound series is small. On the other hand, if the removal of the parameter shuffles this ordering, then that fixed parameter must be regarded as important to the explanation of the overall variation of δ^{3L} . This has been explored in Figure 5, which shows δ^{3L} computed from eq 9 (the complete formula) and its simplified versions using the following fixed average values of the following parameters, each one at a time: Figure 5a energy of the 2PA-active excited state (E_{pf}); Figure 5b detuning factor (E_{det}); Figure 5c the transition dipole moment between the ground and the 1PA-active excited state (μ_{fo}); Figure 5d effective transition dipole moment between the 1PA-active and the 2PA-active excited states (μ_{op}). Clearly, Figure 5d is the one with the least correlation, indicating that effective transition dipole moment between the 1PA-active and the 2PA-active excited states (μ_{op}) is the most important term in eq 9, being therefore mostly responsible for the observed variation in δ^{3L} across the series of compounds.

We proceed now with a detailed analysis of the four classes of compounds studied in the present article. In order to judge if a given substitution is advantageous in terms of producing large values of $\delta_{\max}$, we chose, as our benchmark, the molecule 1, with R equal to phenyl (Figure 6) and we start our analysis by adding substituents to the *meta* or *para* positions of the phenyl groups of the benchmark molecule 1. We chose the calculated substituents from a long list of substituents compiled by Hansch et al.⁵⁹ We also included in our calculations a few other substituents of our own. As a consequence, our calculations involved 54 different substituents. In Table S2 of the Supporting Information, we have the values of $\lambda_{1\text{pa}}$, $\lambda_{2\text{pa}}$, and $\delta_{\max}$ for the molecules with R equal to *meta*- and *para*-substituted phenyl groups. We observed that, for the molecules where we have *meta*-substituted phenyl groups, the values of $\lambda_{1\text{pa}}$ are nearly constant, varying only between 323 and 325 nm. For these molecules, there is a greater variation in the values of $\lambda_{2\text{pa}}$. For the case of the molecules with *para*-substituted phenyl groups, we observed a large dependence of $\lambda_{1\text{pa}}$ and $\lambda_{2\text{pa}}$ on the substituent.

The molecules in Supporting Information Table S2 can be arranged in 54 pairs; each pair containing one molecule with *meta*-substituted phenyl groups, and one molecule with the corresponding *para*-substituted phenyl groups. In Figure 7, we plot the calculated values of $\delta_{\max}$ for the molecules with *para*-substituted phenyl groups, versus the values of $\delta_{\max}$ calculated for the molecules with the corresponding *meta*-substituted phenyl groups. We can observe in Figure 7 that, for the majority of cases, the value of $\delta_{\max}$ for a molecule with *para*-substituted phenyl groups is larger than the value of $\delta_{\max}$ for the molecule with the corresponding *meta*-substituted phenyl groups—they lie above the dashed straight line of slope one. We can also observe that, in the five cases where this rule breaks down, the values of $\delta_{\max}$ for these molecules are smaller than the value of $\delta_{\max}$ for the benchmark molecule 1. As a matter of fact, only three molecules with *meta*-substituted phenyl groups have values of $\delta_{\max}$ larger than the benchmark molecule 1. On average, the

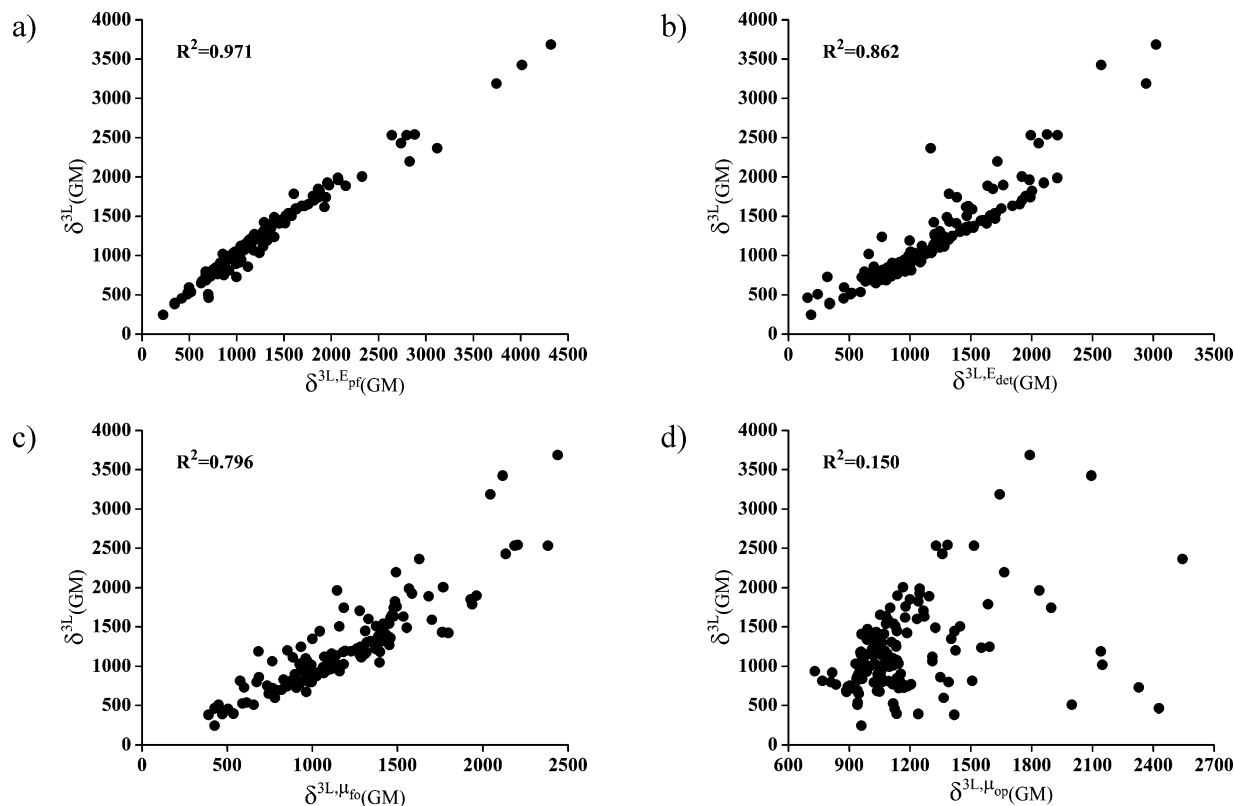


Figure 5. Comparison between the 2PA cross sections, δ^{3L} , obtained using the three-level approximation (vertical axis) and the cross sections obtained from eq 9 keeping one of its four parameters constant at its average value (horizontal axis) for all the molecules studied in present work. The parameters kept constant are (a) E_{pr} , (b) E_{et} , (c) μ_{eo} , and (d) μ_{op} . (1 GM = 10^{-50} cm⁴ s/photon).

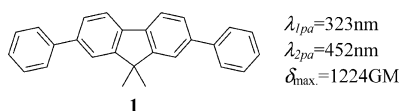


Figure 6. Structure of the molecule chosen as our benchmark to judge if a given substituent is advantageous in terms of producing large values of δ_{max} . The calculated values of λ_{1pa} , λ_{2pa} , and δ_{max} are also indicated. (1 GM = 10^{-50} cm⁴ s/photon).

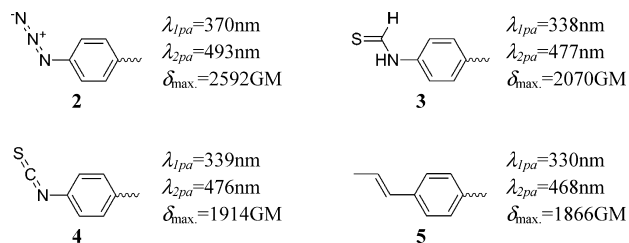


Figure 8. Structures of the *para*-substituted phenyl groups that resulted in the largest calculated values of δ_{max} for this class of molecules. The wavy bond is the position where the group connects to the fluorene unit. Also identified are the calculated values of λ_{1pa} and λ_{2pa} for these molecules. (1 GM = 10^{-50} cm⁴ s/photon.)

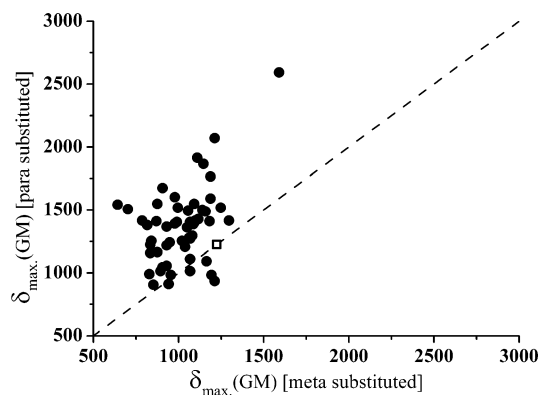


Figure 7. Calculated values of δ_{max} for the molecules with *para*-substituted phenyl groups versus the values of δ_{max} for the molecules with the corresponding *meta*-substituted phenyl groups. This graph contains results for 54 pairs of molecules. The dashed line has slope one and is present only to classify the molecules in two groups. The open square represents the value of δ_{max} for the benchmark molecule 1. (1 GM = 10^{-50} cm⁴ s/photon.)

values of δ_{max} for the molecules with *meta*-substituted phenyl groups are 23% smaller than the molecules with *para*-substituted phenyl groups. Therefore, we have that, when choosing the positions of the substituents of the phenyl groups, it is much more advantageous to place the substituents in the *para*

positions. This result, that *para*-substituted phenyl groups are much more efficient in increasing the values of δ_{max} than *meta*-substituted phenyl groups, can be attributed to the well-known fact that groups that are positioned *meta* to each other are not in mutual conjugation as is the case of groups positioned *para* to each other.^{60–62} This effect has also been described in the case of the 2PA cross sections.⁶³

In Figure 8, we have the four *para*-substituted phenyl groups that result in the largest calculated values of δ_{max} for this class of molecules. There are three possible effects that can contribute to the values of δ_{max} for a molecule: (a) steric effects that break the conjugated path of the molecule, (b) the donor/acceptor ability of the groups linked to the conjugated bridge, and (c) the length of the conjugated path of the molecule. The influence that these three effects have in the 2PA cross sections of organic molecules is described in a recent review by Pawlicki et al.⁶⁴ In the case of the molecules in Supporting Information Table S2, the dihedral angles between the substituted phenyl groups and the central fluorene unit are all around $50 \pm 1^\circ$. Therefore, the differences between the values of δ_{max} for these molecules

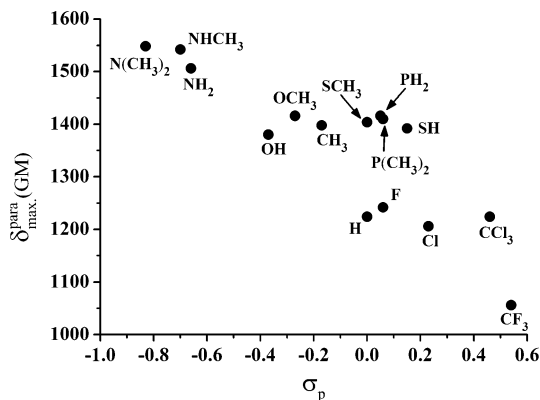


Figure 9. Calculated values of $\delta_{\max}$ for the molecules with *para*-substituted phenyl groups with less than two conjugated atoms in the substituent versus the Hammett's σ_p values of the substituents. (1 GM = 10^{-50} cm⁴ s/photon.)

cannot be attributed to steric effects breaking the conjugated path between the substituted phenyl groups and the fluorene unit. Had we also considered *ortho*-substituted phenyl groups in our calculations, such steric effects would certainly play a role, most likely disrupting the conjugated path, thus resulting in decreased values of $\delta_{\max}$.

By analyzing the values of $\delta_{\max}$ in Supporting Information Table S2 for the molecules containing *para*-substituted phenyl groups, one can observe that the largest values are obtained for substituents with a conjugated path extending over more than one atom. Moreover, when analyzing the effects due to both the donor/acceptor ability and the length of the conjugated path of the *para*-substituted phenyl groups in Supporting Information Table S2, we cannot easily uncouple these effects, but we can, in some cases, even out the latter by considering substituents with similar conjugation paths. In Figure 9, we plot the calculated values of $\delta_{\max}$ for the molecules with *para*-substituted phenyl groups with less than two conjugated atoms in the substituent, versus the Hammett's σ_p values of the substituents. We can observe that, as we proceed from strong donor substituents to strong acceptor substituents, as the value of σ_p increases, the value of $\delta_{\max}$ decreases. We have, for example, that, when we put a strong donor substituent like the dimethylamino (σ_p equal to -0.83) in the *para* position of the phenyl group, we obtain a calculated $\delta_{\max}$ equal to 1548 GM. With an accepting substituent like trifluoromethyl (σ_p equal to 0.54), we obtain a smaller $\delta_{\max}$ equal to 1056 GM. A similar analysis performed for the molecules with substituents with two or more conjugated atoms shows a similar trend, but the points in the plot are more scattered. Our results indicate that even though the nitro substituent (σ_p equal to 0.78) shows electron delocalization, the calculated value of $\delta_{\max}$ is equal to 906 GM, the smallest value for the molecules of this series. This result, that bis-donor substituents perform better than bis-acceptor substituents when comes to increasing the value of the 2PA cross section in fluorene derivatives, has been observed before.⁶⁵ Therefore, our results confirm that it is much more advantageous to have donor groups connected to the fluorene core than to have acceptors.

An apparent counterexample for the rule established above is the case of the molecules with the azido (molecule 2) and the isothiocyanato (molecule 4) substituents in the *para* position. Even though these substituents have σ_p equal to, respectively, 0.08 and 0.38, these two molecules have values of $\delta_{\max}$ equal to, respectively, 2592 GM and 1914 GM, two of the largest values of $\delta_{\max}$ for the molecules of this series. These values of

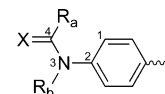


Figure 10. Structure of the groups chosen to exemplify the effect of the rotation of the bond between the nitrogen and the phenyl ring (d_{1234}) in the calculated value of $\delta_{\max}$. The position X can be equal to O (amides) or S (thioamides). The positions R_a and R_b were systematically substituted by hydrogen and methyl. The wavy bond is the position where the group connects to the fluorene unit.

TABLE 2: Calculated Values of λ_{1pa} , λ_{2pa} , and $\delta_{\max}$ and of the rotation (in Degrees) of the Bond between the Nitrogen and the Phenyl Ring (d_{1234}) that Serves as an Indication of Conjugation, for the Amides (X = O) and Thioamides (X = S)^a

	X = O			
	d_{1234}	λ_{1pa} (nm)	λ_{2pa} (nm)	$\delta_{\max}$ (GM)
$R_a = H$ and $R_b = H$	1	330	471	1674
$R_a = CH_3$ and $R_b = H$	39	328	467	1518
$R_a = H$ and $R_b = CH_3$	48	329	466	1494
$R_a = CH_3$ and $R_b = CH_3$	62	328	466	1518
	X = S			
	d_{1234}	λ_{1pa} (nm)	λ_{2pa} (nm)	$\delta_{\max}$ (GM)
$R_a = H$ and $R_b = H$	1	338	477	2070
$R_a = CH_3$ and $R_b = H$	55	330	468	1488
$R_a = H$ and $R_b = CH_3$	49	331	471	1416
$R_a = CH_3$ and $R_b = CH_3$	63	330	467	1500

^a The variation of d_{1234} is a consequence of the substitution of hydrogen by methyl in positions R_a and R_b (see Figure 10). (1 GM = 10^{-50} cm⁴ s/photon).

$\delta_{\max}$ should be compared with the result obtained for the molecule with the dimethylamino substituent. Our results indicate that these two molecules, which possess accepting substituents, have values of $\delta_{\max}$ considerably larger than the molecule with a strong donor substituent. Therefore, we have advanced that the values of $\delta_{\max}$ for molecules 2 and 4 are larger due to a longer conjugated path. This longer conjugated path is also reflected in the longer wavelengths of the 1PA and 2PA transitions of these two molecules. It is expected that groups that are longer than the ones studied here might yield values of $\delta_{\max}$ even larger than the ones obtained for molecule 2.

An example of the effect that steric factors can have in the calculated values of $\delta_{\max}$ is provided by the amides and thioamides represented in Figure 10. In this case, the calculated values of $\delta_{\max}$ are affected by the rotation of the bond between the nitrogen and the phenyl ring. We can observe in Table 2 that, when we replace hydrogen by methyl in positions R_a and R_b , the RM1 optimized torsion angle d_{1234} , that serves as an indication of conjugation, increases. This considerable increase in d_{1234} results from the steric interaction between the added methyl group and the phenyl ring. This increase in d_{1234} , and consequent disruption in conjugation, is accompanied by a decrease in the calculated value of $\delta_{\max}$. This effect is much more accentuated in the case of the thioamides.

For the next series of calculations, we chose molecules where the groups R connected to the fluorene unit (Figure 1) are six-membered heterocycles. The groups chosen were pyridine, pyridazine, pyrimidine, pyrazine, and 1,3,5-triazine. Connecting these five groups in all possible arrangements to the fluorene unit, we generated a series of ten different molecules. In Table S3 of the Supporting Information, we have the calculated values of λ_{1pa} , λ_{2pa} , and $\delta_{\max}$ for these molecules. When comparing the positions of the 1PA and 2PA transitions for this series of

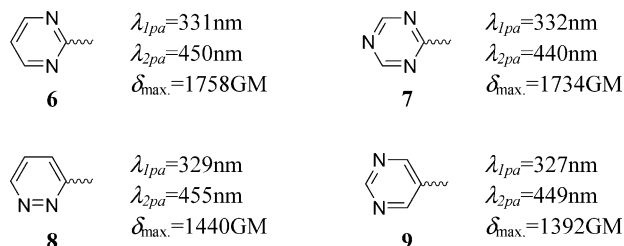


Figure 11. Structure of the four six-membered heterocyclic groups that resulted in values of $\delta_{\text{max.}}$ larger than the benchmark molecule 1. The wavy bond is the position where the group connects to the fluorene unit. Also identified are the calculated values of λ_{1pa} and λ_{2pa} for the resulting molecules. (1 GM = 10^{-50} cm⁴ s/photon).

molecules, it is noteworthy that the molecule containing the 4-pyridinyl group is the one where the transitions happen at the shortest wavelengths, particularly its 1PA transition that is blue-shifted 26 nm with respect to the benchmark molecule 1. We can observe in Table S3 that, of the ten six-membered heterocyclic arrangements studied, only four groups resulted in molecules with values of $\delta_{\text{max.}}$ larger than the benchmark molecule 1. In Figure 11, we have the structure of these four groups.

We can observe in Table S3 that neither one of the three arrangements with pyridinyl groups resulted in values of $\delta_{\text{max.}}$ greater than the one obtained for the benchmark molecule 1; and that, of the three, the 4-pyridinyl group resulted in the smallest value of $\delta_{\text{max.}}$. Our results indicate that the values of λ_{1pa} , λ_{2pa} , and $\delta_{\text{max.}}$ for the molecules containing the 2-pyridinyl and 3-pyridinyl groups are nearly identical. The 2-pyrazinyl group is not advantageous either, because the resulting molecule has a $\delta_{\text{max.}}$ equal to only 906 GM. The groups derived from pyridazine show an interesting behavior, as the value of $\delta_{\text{max.}}$ obtained for the molecule containing the 3-pyridazinyl group (molecule 8) is nearly twice as large as the one obtained from the 4-pyridazinyl group ($\delta_{\text{max.}}$ equal to 768 GM). The 2PA transition for the molecule with the 4-pyridazinyl group is also blue-shifted 15 nm with respect to molecule 8. In the case of the three arrangements with pyrimidinyl groups, our results indicate that the resulting molecules have values for the absorption wavelengths λ_{1pa} and λ_{2pa} that do not vary much between molecules (only 5 nm separate the shortest wavelength from the longest). However, there is a substantial variation in the values of $\delta_{\text{max.}}$ depending on the position of the connection between the pyrimidinyl group and the fluorene unit. The molecule obtained from the 4-pyrimidinyl group has a $\delta_{\text{max.}}$ of only 744 GM, while the molecules obtained from the 5-pyrimidinyl (molecule 9) and the 2-pyrimidinyl (molecule 6) groups have much larger values of $\delta_{\text{max.}}$. The molecule resulting from the 1,3,5-triazine group (molecule 7) has a value of $\delta_{\text{max.}}$ comparable to the one obtained for the best molecule of this series (molecule 6).

The molecules containing six-membered heterocyclic groups considered in this study have a conjugated path length equal to the one of the benchmark molecule 1. Note that there are no relevant steric effects in these molecules. Therefore, the differences in the calculated values of $\delta_{\text{max.}}$ can be attributed to differences in the donor/acceptor abilities of the groups.

Even though there are four promising molecules among the ones derived from six-membered heterocycles (molecules 6 through 9), their values of $\delta_{\text{max.}}$ are still smaller than the ones obtained for the molecules containing the most promising *para*-substituted phenyl groups (molecules 2 through 5).

The next series of calculations involved five-membered heterocyclic groups connected to the fluorene unit. We consid-

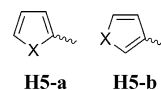


Figure 12. Structures of the two subtypes of five-membered heterocyclic groups studied in the present work. The position X can be equal to NCH₃, O, or S. Also studied were the groups where one of the three carbons of the ring was replaced by nitrogen. The wavy bond is the position where the group connects to the fluorene unit.

TABLE 3: Average Values of λ_{1pa} , λ_{2pa} , and $\delta_{\text{max.}}$ Obtained for the Molecules Containing Five-Membered Heterocyclic Groups as Functions of the Subtype of the Group as Well as of the Heteroatom X (Figure 12)^a

	subtype		X		
	H5-a	H5-b	NCH ₃	O	S
λ_{1pa} (nm)	348	334	338	343	341
λ_{2pa} (nm)	490	464	479	472	479
$\delta_{\text{max.}}$ (GM)	1638	1338	1218	1926	1320

^a 1 GM = 10^{-50} cm⁴ s/photon.

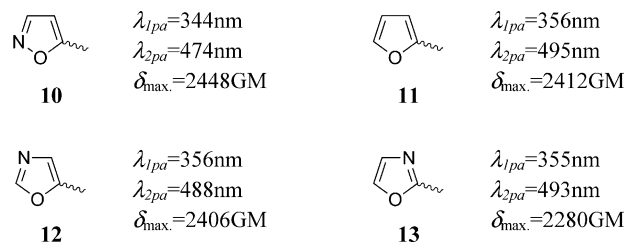


Figure 13. Structure of the four five-membered heterocyclic groups that resulted in the largest calculated values of $\delta_{\text{max.}}$ for this class of molecules. The wavy bond is the position where the group connects to the fluorene unit. Also identified are the calculated values of λ_{1pa} and λ_{2pa} for the resulting molecules. (1 GM = 10^{-50} cm⁴ s/photon.)

ered groups derived from *N*-methyl-pyrrole, furan, thiophene, *N*-methyl-pyrazole, *N*-methyl-imidazole, oxazole, isoxazole, thiazole, and isothiazole. We calculated a total of 24 different molecules containing these groups. In Table S4 of the Supporting Information, we have the values of λ_{1pa} , λ_{2pa} , and $\delta_{\text{max.}}$ for these molecules. To facilitate the analysis that follows, we classified the groups into two subtypes (see Figure 12).

In Table 3, we have the average values of λ_{1pa} , λ_{2pa} , and $\delta_{\text{max.}}$ for this series of molecules as a function of the heteroatom in position X and of the subtype of the group. We can observe in Table 3 that the average values of λ_{1pa} and λ_{2pa} are more susceptible to changes in the subtype of the group than to changes in the heteroatom X. Our results indicate that molecules containing groups of the subtype H5-a absorb light at longer wavelengths than molecules of the subtype H5-b and that it is λ_{2pa} that varies the most. The average values of $\delta_{\text{max.}}$, on the other hand, are susceptible to both the subtype of the group and the heteroatom in position X of the ring. Our results indicate that, for molecules with heterocyclic groups of subtype H5-a, the values of $\delta_{\text{max.}}$ are, on average, 300 GM larger than those of molecules with heterocyclic groups of subtype H5-b. Our results further indicate that molecules with X equal to O show the largest average value of $\delta_{\text{max.}}$. In Figure 13, we have the four five-membered heterocyclic groups that resulted in the largest calculated values of $\delta_{\text{max.}}$ for this series of molecules. In conformity with the average values in Table 3, we have shown that the most promising molecules possess the groups furanyl, oxazolyl, and isoxazolyl of subtype H5-a.

In Table 4, we examine in more detail the average values of $\delta_{\text{max.}}$ obtained for the five-membered heterocyclic groups as a function of the subtype of the group and of the heteroatom in position X, both taken simultaneously. We can observe the

TABLE 4: Average Values of $\delta_{\max}$ and of the Torsion Angles between the Heterocycles and the Fluorene Unit Obtained for the Molecules Containing Five-Membered Heterocyclic Groups as a Function of Both the Heteroatom X and the Subtype of the Group Taken Simultaneously (Figure 12)^a

X	subtype	$\delta_{\max}$ (GM)	torsion (degrees)
NCH ₃	H5-a	852	50
	H5-b	1584	31
O	H5-a	2388	16
	H5-b	1464	28
S	H5-a	1680	40
	H5-b	960	38

^a (1 GM = 10^{-50} cm⁴ s/photon.)

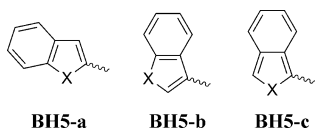


Figure 14. Structures of the three subtypes of benzo derivatives of five-membered heterocyclic groups studied in the present work. The position X can be equal to NCH₃, O, or S. Also studied were the groups where the available carbon atom of the five-membered ring was replaced by nitrogen. The wavy bond is the position where the group connects to the fluorene unit.

interesting behavior that, while for X equal to NCH₃ the molecules with groups of subtype H5-a have, on average, values of $\delta_{\max}$ smaller than the molecules with groups of subtype H5-b; in the cases where we have X equal to O or S, the converse is true. The length of the conjugated path is the same for all groups considered. Therefore, the observed differences can only be explained by a combination of the donor/acceptor ability of the groups with steric effects. In Table 4, we also have the average values of the RM1 optimized torsion angle between the heterocycles and the fluorene unit. We have shown that, in the case of groups with X equal to NCH₃, the average value of the torsion angles for groups of subtype H5-a are considerably larger than for groups of subtype H5-b. This means that the methyl substituent bonded to the nitrogen of groups of subtype H5-a interact sterically with the fluorene unit and disrupt the conjugated path of the molecule by decreasing the values of $\delta_{\max}$. In the cases where we have X equal to S, the average torsion angles are nearly the same for groups of both subtypes and the differences in the average values of $\delta_{\max}$ are due to differences in the donor/acceptor abilities of the groups. This means that the groups of subtype H5-a are electronically more favorable than groups of subtype H5-b. When we have X equal to O, the strong steric effects observed when we have X equal to NCH₃ are not present, and the torsion angles for groups of subtype H5-a are smaller than for groups of subtype H5-b. The lack of detrimental steric effects, together with more favorable electronic effects for groups of subtype H5-a results in larger values of $\delta_{\max}$ for these molecules.

Even though the five-membered heterocyclic groups are structurally different from the phenyl group in the benchmark molecule 1, we still have the same number of π electrons in both cases. Therefore, results obtained for the molecules with five-membered heterocyclic groups can be reasonably compared to those for the benchmark molecule 1. Our results indicate that the values of $\delta_{\max}$ shown in Figure 13 are considerably larger than that for the benchmark molecule 1. However, molecules 10 through 13 still have values of $\delta_{\max}$ smaller than that for molecule 2.

TABLE 5: Average Values of λ_{1pa} , λ_{2pa} , and $\delta_{\max}$ Obtained for Molecules Containing Groups Derived from Benzo Derivatives of Five-Membered Heterocycles as Functions of the Subtype of the Group, as Well as of the Heteroatom X (Figure 14)^a

	subtype			X		
	BH5-a	BH5-b	BH5-c	NCH ₃	O	S
λ_{1pa} (nm)	355	333	405	362	363	369
λ_{2pa} (nm)	498	470	541	511	504	494
$\delta_{\max}$ (GM)	2232	762	1278	936	2082	1248

^a (1 GM = 10^{-50} cm⁴ s/photon.)

For our next series of calculations, we chose molecules with groups derived from benzo derivatives of five-membered heterocycles that connect to the fluorene unit through the five-membered ring. In analogy with what has been done with the five-membered heterocycles described above, we classified the groups into three subtypes (see Figure 14). We considered groups where the position X was substituted by NCH₃, O, or S, and also studied the substitution of the available carbon atom of the five-membered ring by nitrogen. We calculated a total of 18 different molecules containing these groups. In Table S5 of the Supporting Information, we have the values of λ_{1pa} , λ_{2pa} , and $\delta_{\max}$ for these molecules.

In Table 5, we have the average values of λ_{1pa} , λ_{2pa} and $\delta_{\max}$ for this series of molecules as a function of the heteroatom in position X and the subtype of the group. We can observe in Table 5 that, similarly to what was observed in Table 3 for the five-membered heterocyclic groups, the average values of λ_{1pa} and λ_{2pa} are not very susceptible to changes in the heteroatom X. On the other hand, the average values of λ_{1pa} and λ_{2pa} show large variations as a function of the subtype of the group. It is striking how similar the average values of λ_{1pa} and λ_{2pa} for the molecules containing groups of subtypes BH5-a and BH5-b are to the values obtained in Table 3 for the molecules containing five-membered heterocyclic groups of subtypes H5-a and H5-b. The largest variations in λ_{1pa} and λ_{2pa} are observed for molecules containing groups of the subtype BH5-c that show transitions at wavelengths considerably longer than in the case of subtypes BH5-a and BH5-b. Similarly to what was observed for the five-membered heterocyclic groups, the average values of $\delta_{\max}$ for this series are susceptible to both the heteroatom in position X and the subtype of the group. We can observe in Table 5 that molecules with X equal to O have values of $\delta_{\max}$ that are, on average, more than two times larger than for the case of X equal to NCH₃, and that molecules containing groups of subtype BH5-a are, on average, nearly three times more promising than molecules containing groups of subtype BH5-b. Our results further indicate that groups of subtype BH5-c result in molecules with average values of $\delta_{\max}$ that are 954 GM smaller than molecules with groups of subtype BH5-a. The molecules of this series have groups with 10 π electrons and, in principle, have an advantage, in terms of the length of the conjugated path, over the benchmark molecule 1 that has groups with 6 π electrons. However, only six groups of this series result in molecules with values of $\delta_{\max}$ larger than the one for the benchmark molecule 1. These six groups are shown in Figure 15. Our results indicate that, of the six groups, three of them have values of $\delta_{\max}$ that are larger than that for molecule 2 and these three groups are of subtype BH5-a. With values of $\delta_{\max}$ greater than 3000 GM, the most promising groups are the 2-benzoxazolyl and the 2-benzofuranyl. In contrast to the large values of $\delta_{\max}$ obtained from these two groups, this series of molecules also yielded the smallest value of $\delta_{\max}$ obtained in

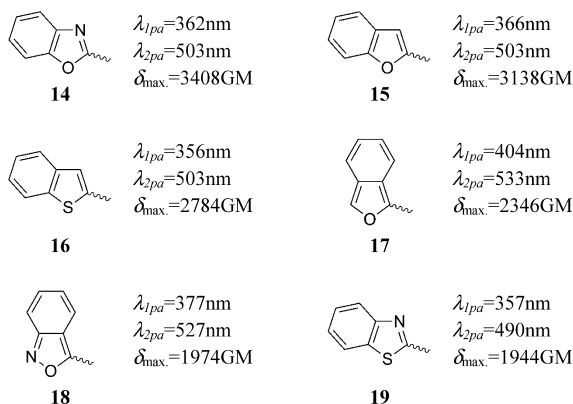


Figure 15. Structure of the six benzo derivatives of five-membered heterocyclic groups that resulted in values of δ_{max} larger than the benchmark molecule 1. The wavy bond is the position where the group connects to the fluorene unit. Also identified are the calculated values of λ_{1pa} and λ_{2pa} for the resulting molecules. (1 GM = 10^{-50} cm⁴ s/photon.)

TABLE 6: Average Values of δ_{max} and of the Torsion Angles between the Heterocyclic Group and the Fluorene Unit Obtained for the Molecules Containing Groups Derived from Benzo Derivatives of Five-Membered Heterocycles as a Function of the Heteroatom X and of the Subtype of the Group, Both Variables Taken Simultaneously (Figure 14)^a

X	Subtype	δ_{max} (GM)	Torsion (degrees)
NCH ₃	BH5-a	1062	49
	BH5-b	864	44
	BH5-c	888	50
O	BH5-a	3276	20
	BH5-b	816	43
	BH5-c	2160	37
S	BH5-a	2364	39
	BH5-b	612	52
	BH5-c	780	54

^a (1 GM = 10^{-50} cm⁴ s/photon.)

our calculations. The group of the subtype BH5-b derived from thiophene resulted in a value of δ_{max} of only 516 GM.

In Table 6, we have a detailed analysis of the average values of δ_{max} for this class of molecules as a function of both the heteroatom in position X and the subtype of the group, taken simultaneously. In Table 6, we also have the average value of the RM1 optimized torsion angle between the heterocycles and the fluorene unit. Our results indicate that, in congruence with the results shown in Figure 15, the most promising values were obtained for molecules with groups of subtype BH5-a with X equal to O or S, and for molecules with groups of subtype BH5-c, with X equal to O. Our results indicate that these most promising values were obtained for the molecules with the smallest values of the torsion angle. In the case where X is equal to NCH₃, we can still observe some steric effects in the case of groups of subtype BH5-a and BH5-c, but these effects play a minor role in the values of δ_{max} .

The analysis above shows that several molecules containing *para*-substituted phenyl groups connected to the fluorene unit are very promising candidates as molecules with large values of δ_{max} . As a general rule, the best molecules of this class possess a longer conjugated path. The molecules containing *meta*-substituted phenyl groups are not as interesting as the ones containing *para*-substituted phenyl groups. The study of molecules containing heterocyclic groups showed that it is more interesting to have five-membered heterocyclic groups of subtype H5-a or their benzo derivatives of subtype BH5-a

connected to the fluorene unit. Among those, the best molecules contained X equal to O in the heterocycle. This study also showed that, due to steric factors, it is better to avoid having the group X equal to NCH₃ in the heterocyclic group of subtype H5-a.

4. Conclusions

In this article, we employed the computational procedure previously described by us⁴⁶ to theoretically study the two-photon absorption parameters (λ_{2pa} and δ_{max}) of a series of 161 different molecules derived from a symmetrically substituted fluorene unit. The groups considered were *meta*- or *para*-substituted phenyl groups, five- and six-membered heterocycles, and benzo derivatives of five-membered heterocycles.

We also show that the three-level approximation has to be adapted to include the effects of a fourth 2PA excited state in order to properly rationalize the values of δ_{max} of the substituted fluorenes. We also show that the parameter mostly responsible for the large observed variation on the calculated values of δ_{max} for the studied molecules is the effective transition dipole moment between the 1PA-active and the two 2PA-active excited states.

We confirm that, in the case of the molecules containing substituted phenyl groups, it is more advantageous to have the substituents in the *para* position than to have them in the *meta* position. We also confirm that molecules containing donating substituents are more interesting for further investigation than molecules containing accepting substituents. We further confirm that molecules containing groups with a longer conjugated path have larger values of δ_{max} .

Based on our calculations, we propose that molecules containing *para*-azido-phenyl groups should be considered as candidate synthetic targets for molecules with large values of δ_{max} . Our study of molecules containing five- and six-membered heterocycles further indicates that molecules containing groups such as 5-isoxazolyl, 2-furanyl and 5-oxazolyl are good candidates for molecules with large values of δ_{max} . From our computational study, the most auspicious molecules were the ones obtained from benzo derivatives of five-membered heterocycles, of which the two most promising ones are 2,2'-(9,9-dialkyl-9H-fluorene-2,7-diyl)dibenzo[d]oxazole (molecule 14, Figure 15) and 2,2'-(9,9-dialkyl-9H-fluorene-2,7-diyl)dibenzo-furan (molecule 15, Figure 15).

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Supporting Information Available: Tables containing the calculated values of λ_{1pa} , λ_{2pa} , and δ_{max} , as well as results for the three-level approximation analysis for the 161 molecules studied. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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